Chapter 3

Parallel and Perpendicular Lines

Unit 1 — Lines and Angles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Parallel lines	Coplanar lines that never intersect ($\ell \parallel m$)
Perpendicular lines	Lines that intersect at 90° ($\ell \perp m$)
Skew lines	Non-coplanar, non-intersecting lines
Transversal	A line that crosses two or more other lines
Corresponding angles	Same position at each intersection
Alternate interior angles	Between parallel lines, on opposite sides of transversal
Co-interior (same-side) angles	Between parallel lines, on same side of transversal

Angle Pairs with a Transversal

```
       t
       |
——1—2——  ← line ℓ
——3—4——
       |
——5—6——  ← line m
——7—8——
       |
```

Pair	Angles	Parallel →
Corresponding	∠1 & ∠5, ∠2 & ∠6, etc.	Congruent
Alt. Interior	∠3 & ∠6, ∠4 & ∠5	Congruent
Alt. Exterior	∠1 & ∠8, ∠2 & ∠7	Congruent
Co-interior	∠3 & ∠5, ∠4 & ∠6	Supplementary

Postulate 3-1: Corresponding Angles Postulate

> If two **parallel** lines are cut by a transversal, then **corresponding angles are congruent**.

$$\ell \parallel m \Rightarrow \angle 1 \cong \angle 5$$

This is the foundation for all parallel line angle theorems.

Theorem 3-1: Alternate Interior Angles

> If two **parallel** lines are cut by a transversal, then **alternate interior angles are congruent**.

$$\ell \parallel m \Rightarrow \angle 3 \cong \angle 6$$

Proof basis: Corresponding Angles Postulate + Vertical Angles Theorem.

Theorem 3-2: Consecutive Interior Angles (Co-interior)

> If two **parallel** lines are cut by a transversal, then **consecutive interior angles are supplementary**.

$$\ell \parallel m \Rightarrow m\angle 3 + m\angle 5 = 180^\circ$$

Also called **co-interior** or **same-side interior** angles.

Theorem 3-3: Alternate Exterior Angles

> If two **parallel** lines are cut by a transversal, then **alternate exterior angles are congruent**.

$$\ell \parallel m \Rightarrow \angle 1 \cong \angle 8$$

Theorem 3-4: Perpendicular Transversal

> In a plane, if a line is perpendicular to one of two **parallel** lines, it is perpendicular to the other.

$$\ell \parallel m \text{ and } t \perp \ell \Rightarrow t \perp m$$

Converses — Proving Lines Parallel

The converses are used to **prove** lines are parallel:

Theorem	Condition	Conclusion
Post. 3-2	Corresponding $\angle$s $\cong$	Lines $\parallel$
Thm. 3-5	Alt. interior $\angle$s $\cong$	Lines $\parallel$
Thm. 3-6	Consec. interior $\angle$s supp.	Lines $\parallel$
Thm. 3-7	Alt. exterior $\angle$s $\cong$	Lines $\parallel$

Postulate 3-2: Converse of Corresponding Angles

If two lines are cut by a transversal so that **corresponding angles are congruent**, then the lines are **parallel**.

$$\angle 1 \cong \angle 5 \Rightarrow \ell \parallel m$$

Theorem 3-5: Converse of Alternate Interior Angles

> If two lines are cut by a transversal so that **alternate interior angles are congruent**, then the lines are **parallel**.

$$\angle 3 \cong \angle 6 \Rightarrow \ell \parallel m$$

Theorem 3-6: Converse of Consecutive Interior Angles

If two lines are cut by a transversal so that **consecutive interior angles are supplementary**, then the lines are **parallel**.

$$m\angle 3 + m\angle 5 = 180^\circ \Rightarrow \ell \parallel m$$

Theorem 3-7: Parallel Lines in a Plane

> In a plane, if two lines are both **perpendicular to the same line**, they are **parallel** to each other.

$$t \perp \ell \text{ and } t \perp m \Rightarrow \ell \parallel m$$

Slopes of Parallel and Perpendicular Lines

Relationship	Condition
Parallel lines	Equal slopes: $m_1 = m_2$
Perpendicular lines	Negative reciprocal slopes: $m_1 \cdot m_2 = -1$
Horizontal line	Slope $= 0$
Vertical line	Slope undefined

Slope formula: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

Example — Parallel/Perpendicular Slopes

Line ℓ passes through $(0, 2)$ and $(3, 8)$.

$$m_\ell = \frac{8-2}{3-0} = \frac{6}{3} = 2$$

- **Parallel line** has slope $= 2$
- **Perpendicular line** has slope $= -\frac{1}{2}$

Chapter 3 — Theorem Summary

#	Name	Key Idea
Post. 3-1	Corr. Angles	$\parallel$ lines $\to$ corr. $\angle$s $\cong$
Thm. 3-1	Alt. Interior	$\parallel$ lines $\to$ alt. int. $\angle$s $\cong$
Thm. 3-2	Co-interior	$\parallel$ lines $\to$ consec. int. $\angle$s supp.
Thm. 3-3	Alt. Exterior	$\parallel$ lines $\to$ alt. ext. $\angle$s $\cong$
Thm. 3-4	Perp. Transversal	$t \perp \ell,\ \ell \parallel m \Rightarrow t \perp m$
Post. 3-2	Conv. Corr. Angles	Corr. $\angle$s $\cong$ $\to$ lines $\parallel$
Thm. 3-5	Conv. Alt. Interior	Alt. int. $\cong$ $\to$ lines $\parallel$
Thm. 3-6	Conv. Co-interior	Consec. int. supp. $\to$ lines $\parallel$
Thm. 3-7	Both $\perp$ Same	$t \perp \ell,\ t \perp m \Rightarrow \ell \parallel m$